

MEASURE · REDUCE

COURSE SLIDES · WSQ

Carbon Footprint Management for Sustainability

WSQ Course Code: TGS-2023021099

Conducted by Tertiary Infotech Academy Pte Ltd · UEN 201200696W

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COURSE ADMINISTRATION

Welcome, Outcomes and Assessment

TGS-2023021099 · 2 days · 16 training hours

Digital Attendance (Mandatory)

1

AM, PM and Assessment

Complete every digital attendance event shown by the trainer.

2

Scan and submit

Use your phone camera and verify the submission succeeds.

3

Minimum attendance

At least 75% attendance is required for assessment and funding eligibility.

4

Ask immediately

Tell the trainer if a scan fails; do not leave it unresolved.

About the Trainer



Your Trainer

To be completed by the trainer

NAME

DESIGNATION

QUALIFICATIONS

INDUSTRY EXPERIENCE

CONTACT

About the Trainer



Dr Alfred Ang

Principal Trainer · Tertiary Infotech
Academy

FOCUS

Applied AI, sustainability data and business transformation

ROLE

Course facilitation, assessment briefing and learner support

APPROACH

Evidence-led, practical and decision-focused

ORGANISATION

Tertiary Infotech Academy Pte Ltd

Let's Know Each Other

1

You

Name, organisation and role

2

Experience

Exposure to ESG, operations, energy, finance or reporting

3

Challenge

One footprint or sustainability decision you want to improve

4

Outcome

What a useful carbon-reduction plan would enable

Ground Rules

1

Participate actively and ask questions.

2

Respect different operational perspectives.

3

Use fictionalised or approved business data.

4

Return punctually from breaks.

5

Keep phones silent outside learning tasks.

6

Challenge assumptions with evidence.

Access Your Course Materials

1

LMS

Sign in at lms-tms.tertiaryinfotech.com

2

Course record

Carbon Footprint Management for Sustainability

3

Core files

Slides, Learner Guide and Lesson Plan

4

Activities

One folder per activity with scenario, questions and data

5

Calculator

alfredang.github.io/sgcarboncalculator

6

Assessment

Use only approved open-book materials

Skills Framework Alignment

1

TSC title

Carbon Footprint Management

2

TSC code

PRE-ENV-4001-1.1

3

A1

Analyse carbon emission hotspots

4

A2

Prioritise reduction initiatives

5

A3

Develop reduction plans

6

Knowledge

K1–K9 assessed through nine SAQs

Learning Journey — Two Days

DAY 1

- Measure What Matters — Boundary, Scopes and Hotspots
 - Standards, boundaries and scopes
 - Inventory calculations and data quality
 - Hotspot analysis and economics
 - Activities 1–3

DAY 2

- Move From Insight to Action — Priorities, Plans and Governance
 - Scenarios, technologies and portfolio choices
 - EMS, governance and Singapore context
 - Action planning and capstone
 - Activities 4–9

Learning Outcomes

1

LO1

Analyse carbon emission hotspots using recognised standards, boundaries and reliable activity data.

2

LO2

Prioritise carbon-footprint reduction initiatives using projections, feasibility, cost-benefit and decision tools.

3

LO3

Develop an implementable carbon-footprint reduction plan with governance, targets, monitoring and continual improvement.

Course Map

1

Topic 1 · Analyse Carbon Emission Hotspots

Standards, boundaries, data, scopes and hotspot diagnosis · K1 · K3 · K7 · A1

2

Topic 2 · Prioritise Reduction Initiatives

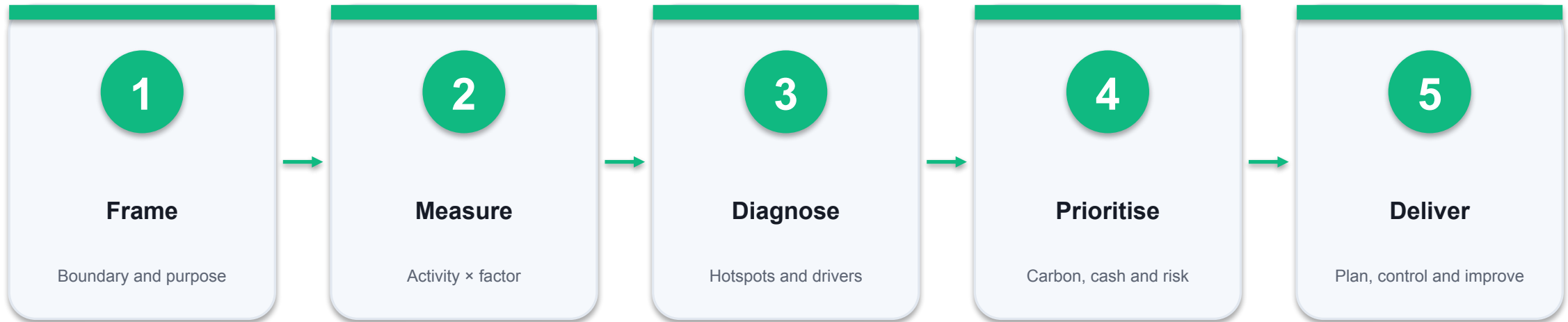
Projection, technology, economics and transparent decisions · K4 · K5 · K8 · A2

3

Topic 3 · Develop the Reduction Plan

EMS governance, action planning, Singapore requirements and review · K2 · K6 · K9 · A3

The Carbon Management Decision Cycle



CORE IDEA

Every activity strengthens one part of the same auditable decision trail.

2 HOURS TOTAL

Assessment Briefing

1

Written Assessment

9 short-answer questions · K1–K9 · 1 hour

2

Case Study

3 applied questions · A1–A3 · 1 hour

3

Open book

Use approved course materials; work independently

4

Evidence

Answer the exact question and show calculations

5

Conduct

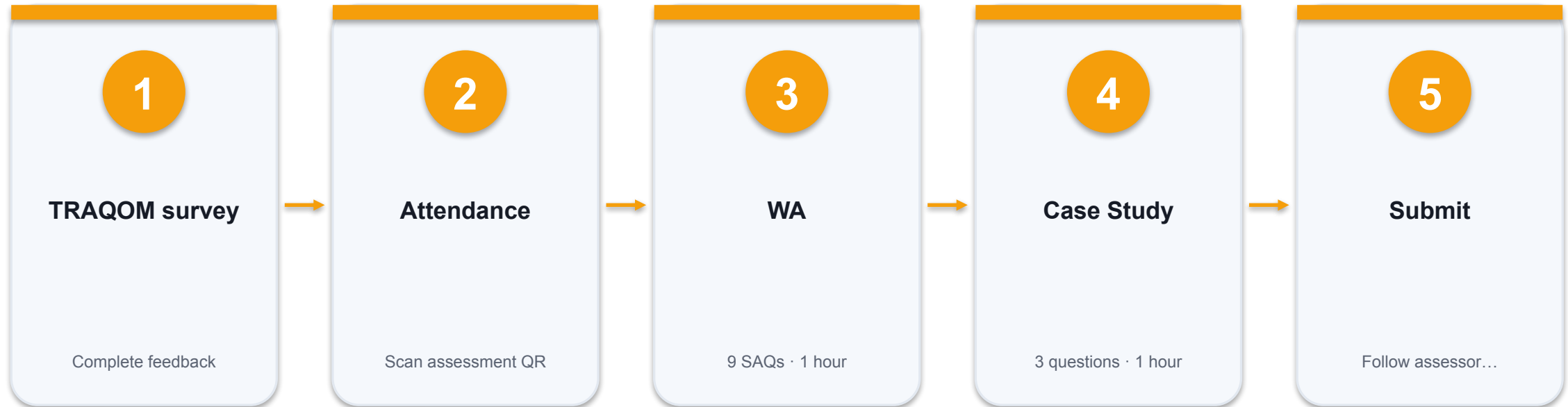
No discussion, photography or recording

6

Outcome

Competent or Not Yet Competent per instrument

Assessment Flow



Course Completion and Funding Requirements

1

Attendance

Minimum 75% based on SSG digital attendance

2

Assessment

Complete both assessment instruments

3

TRAQOM

Complete mandatory course feedback

4

Records

Follow administrative and identity-verification requirements

START WITH PURPOSE

Carbon accounting is not the finish line.

It is the evidence system that helps people choose and deliver better reductions.

TOPIC 1 · K1 · K3 · K7 · A1

Analyse Carbon Emission Hotspots

Standards, boundaries, data, scopes and hotspot diagnosis

A Carbon Inventory Is a Decision System

1

Purpose

Translate physical activity into comparable tCO₂e for decisions.

2

Boundary

State which entities, operations, gases and period are included.

3

Method

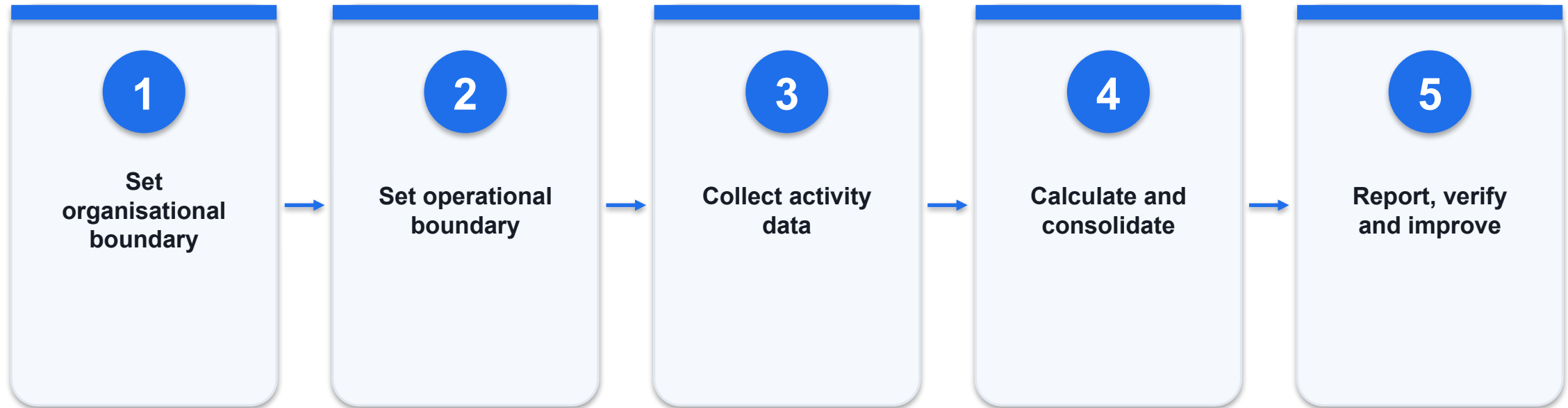
Use consistent activity data, emission factors and global-warming potentials.

4

Evidence

Retain sources, assumptions, controls and recalculation rules.

Build a Corporate Inventory in Five Moves



Five Principles Protect Decision Quality

1

Relevance

Reflect the organisation and users' needs.

2

Completeness

Account for all sources within the boundary; disclose exclusions.

3

Consistency

Enable meaningful comparison over time.

4

Transparency

Disclose methods, assumptions and evidence.

5

Accuracy

Reduce bias and uncertainty as far as practical.

Organisational Boundary Choices

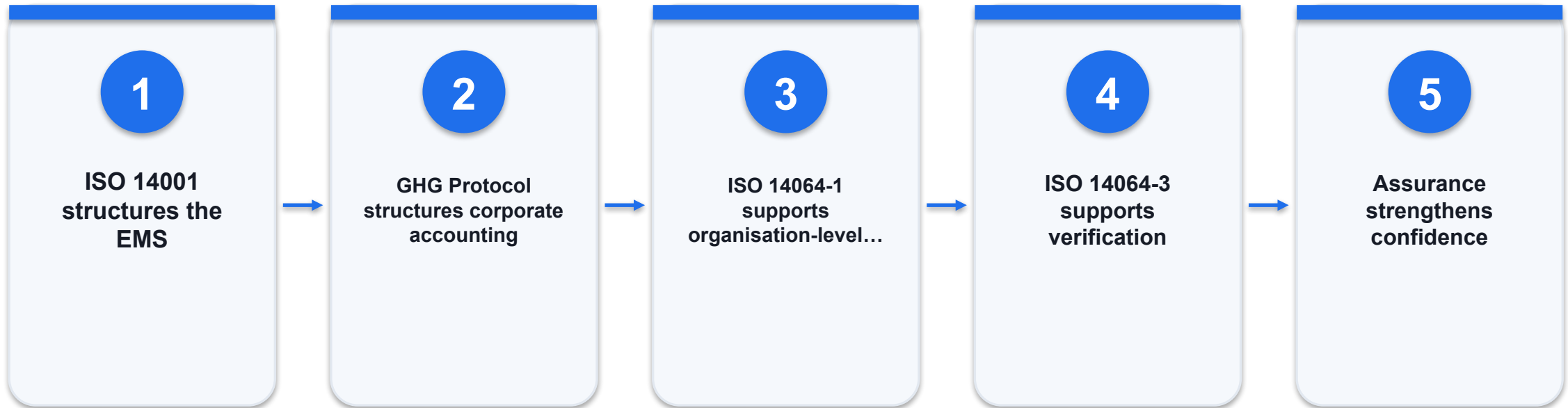
Equity share

- Account according to economic interest
- Useful where ownership shares matter
- Apply consistently across investments

Control approach

- Account for operations under financial or operational control
- Often aligns with management systems
- Document the chosen control test

How the Standards Fit Together



Accounting Standard vs Management Standard

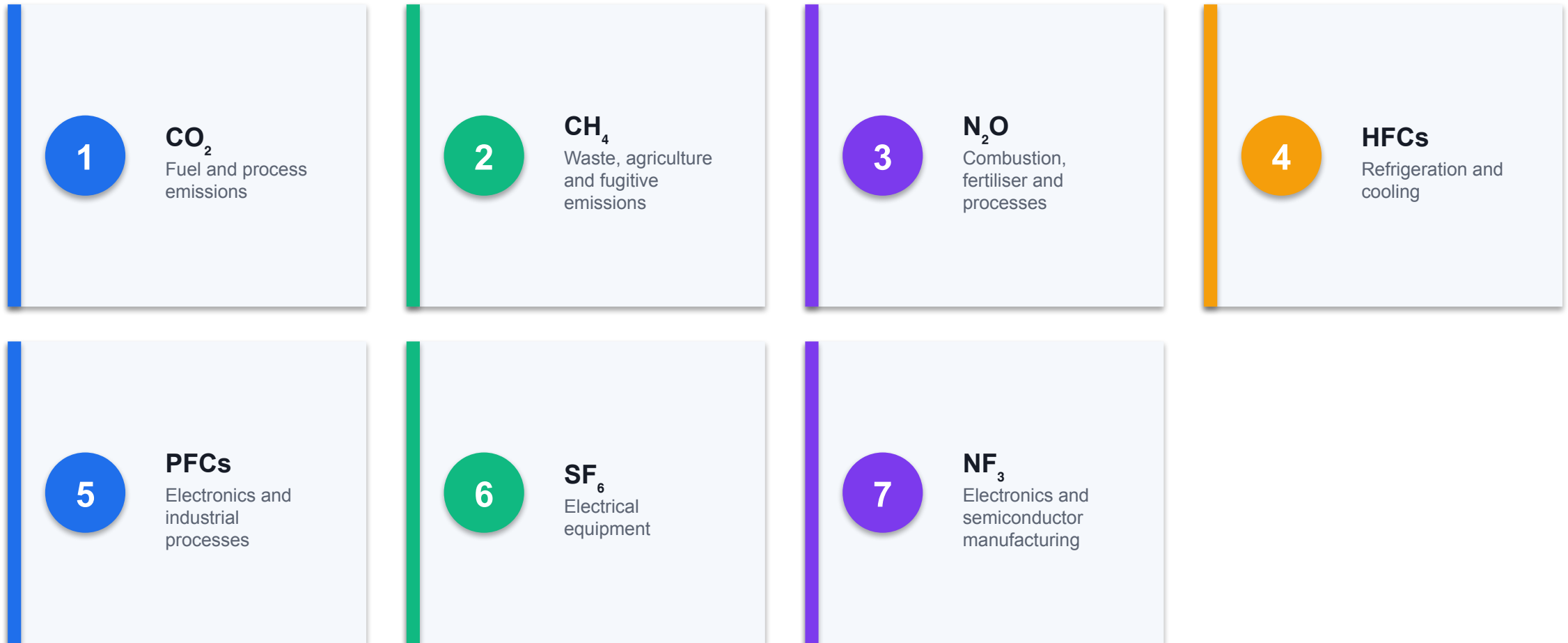
GHG Protocol / ISO 14064

- What and how much is emitted
- Boundaries, categories, calculations
- Inventory quality and reporting

ISO 14001 EMS

- How environmental performance is managed
- Policy, objectives, controls and review
- Continual improvement through PDCA

The Seven Greenhouse Gases



From Activity to tCO₂e

A transparent inventory keeps every assumption visible.

```
Emissions (tCO2e) = Activity data ×  
Emission factor × GWP ÷ unit conversion
```

CHECK 1

Example: kWh × kg CO₂e/kWh ÷ 1,000

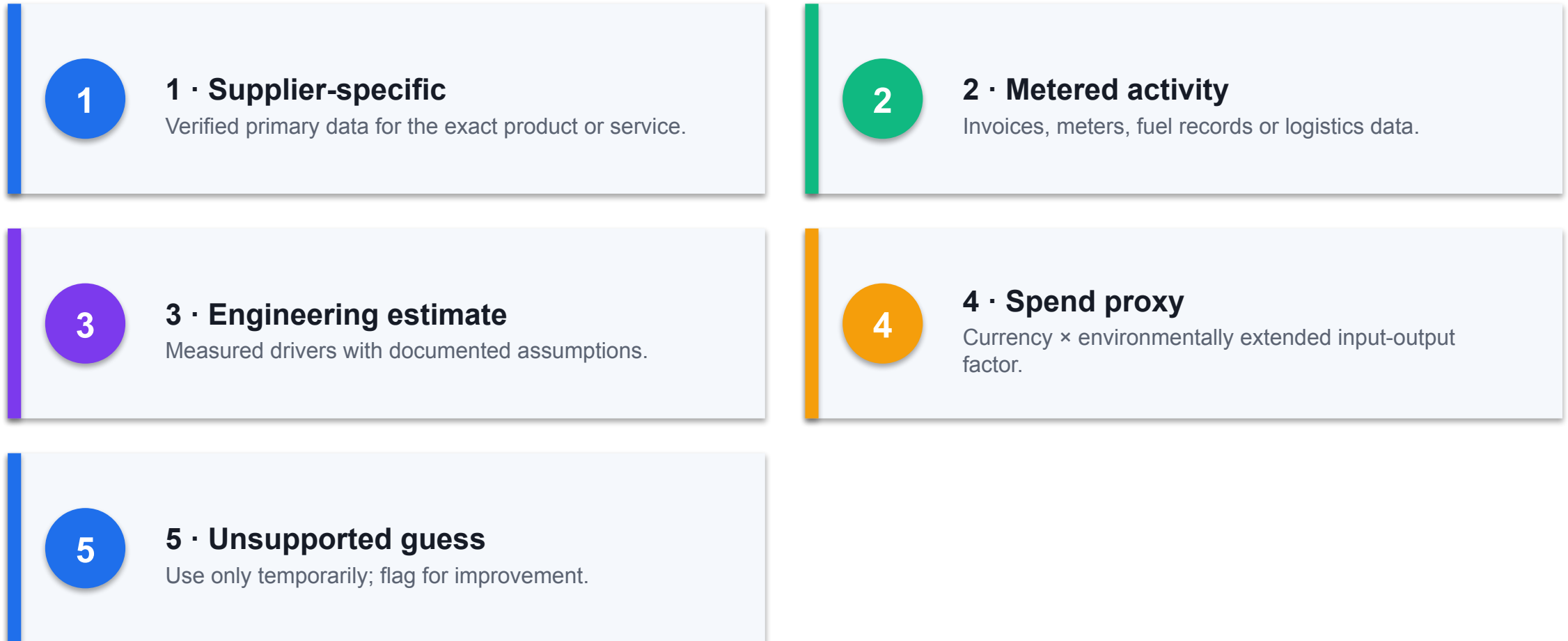
CHECK 2

Use the factor for the correct geography and year

CHECK 3

Record factor source, unit and uncertainty

A Practical Data-Quality Ladder



Three Scopes Prevent Category Confusion

1

Scope 1

Direct emissions from sources owned or controlled.

2

Scope 2

Indirect emissions from purchased electricity, steam, heat or cooling.

3

Scope 3

Other value-chain emissions upstream and downstream.

4

Rule

Scopes describe responsibility for accounting—not moral blame.

Location-Based and Market-Based Views

Location-based

- Average grid emissions where consumption occurs
- Useful for physical system context
- Singapore example: EMA grid factor

Market-based

- Contractual instruments and supplier attributes
- Requires quality criteria and evidence
- Report both where applicable

Follow the Value Chain, Not the Org Chart

1

Upstream

Purchased goods, capital goods, fuels, transport, waste, travel, commuting and leased assets.

2

Downstream

Distribution, processing, use, end-of-life, leased assets, franchises and investments.

3

Screen first

Estimate all relevant categories before investing in precision.

4

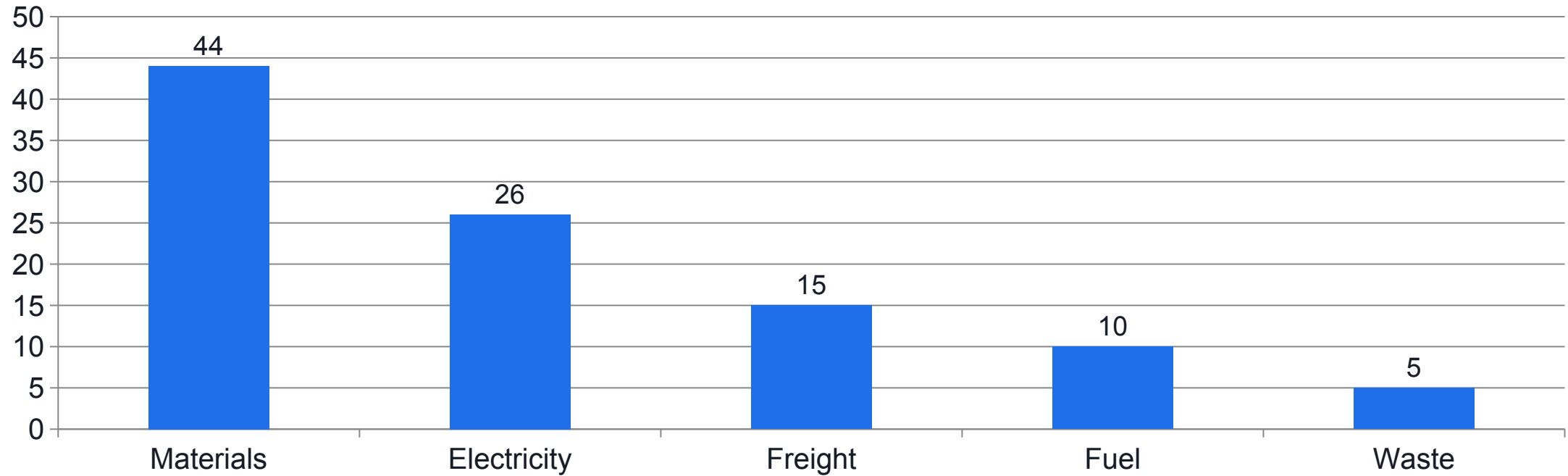
Engage next

Improve supplier and customer data at material hotspots.

The Hotspot Analysis Funnel



Illustrative Footprint by Source



WHAT THE DATA SHOWS

Top two sources represent 70% of the screening footprint; validate them first.

A Hotspot Is More Than the Largest Bar

1

Magnitude

Absolute or intensity-based share of emissions.

2

Influence

Ability to change technology, demand or supplier behaviour.

3

Risk

Carbon price, regulation, disruption or reputation.

4

Strategic fit

Connection to customers, growth and operational priorities.

Common Inventory Failure Modes

1

Missing sources

Completeness check against sites, ledgers and processes.

2

Double counting

Clear ownership, category definitions and consolidation rules.

3

Unit errors

Automated validation and unit conversion controls.

4

Wrong factor

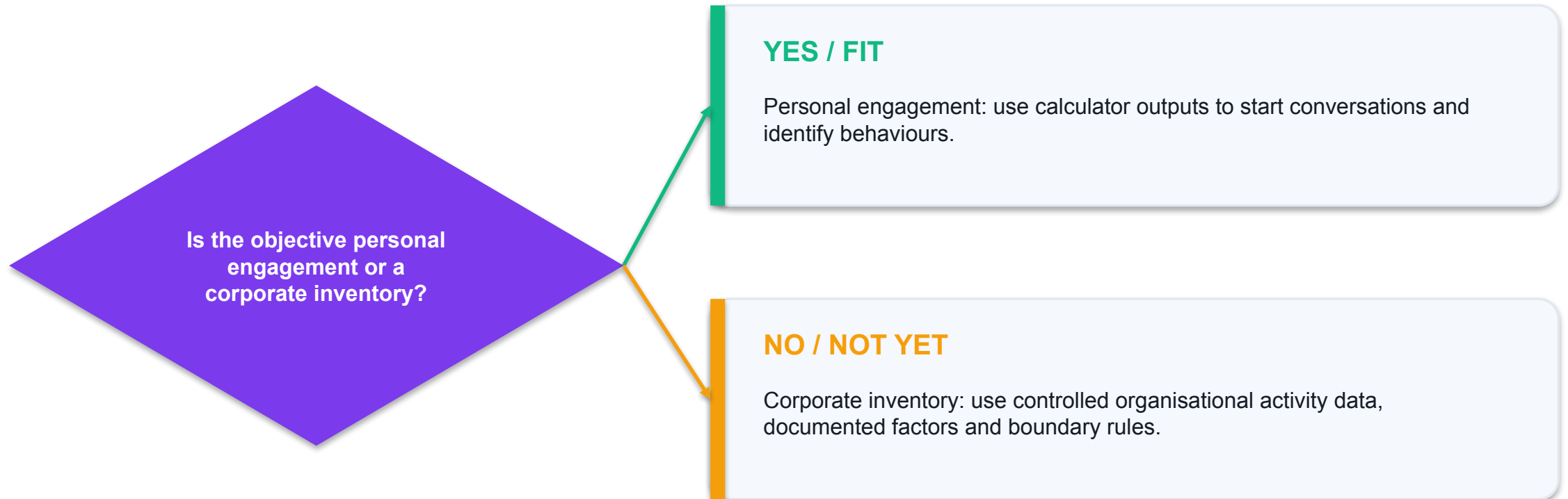
Versioned factor library with geography and year.

5

Silent estimates

Estimate register with owner and replacement date.

Can the SG Carbon Calculator Be Used Here?



The calculator is a screening and learning aid—not a substitute for a GHG Protocol corporate inventory.

Reading Singapore Electricity Data Correctly

1

Published factor

EMA reported 0.402 kg CO₂/kWh for 2024.

2

Correct year

Match the factor year to the reporting period.

3

Correct unit

Convert kg to tonnes only once.

4

Record evidence

Save the factor source and access date.

5

Recalculate

Define when base-year electricity emissions are restated.

Separate Carbon Value From Cash Value

1

Carbon effect

Annual and lifetime tCO₂e avoided.

2

Cash effect

Capex, opex, savings and residual value.

3

Risk effect

Carbon tax, energy volatility and compliance exposure.

4

Strategic effect

Resilience, customer value and learning.

Simple Payback Is Fast—but Incomplete

Useful as a screening metric when cash flows are stable.

```
Simple payback (years) = Initial investment  
÷ Annual net cash saving
```

CHECK 1

Ignores time value of money

CHECK 2

Ignores cash flows after payback

CHECK 3

Never rank long-lived projects by payback alone

NPV Makes Timing Explicit

Discount future net cash flows to present value.

$$\text{NPV} = \sum [\text{Net cash flow}_t \div (1+r)^t] - \text{Initial investment}$$

CHECK 1

Positive NPV creates financial value at the chosen discount rate

CHECK 2

Test energy price, utilisation and maintenance assumptions

CHECK 3

Use sensitivity analysis instead of false precision

Compare Projects per Tonne Avoided

Marginal abatement cost exposes the price of each carbon outcome.

MAC = Lifetime net cost ÷ Lifetime tCO₂e avoided

CHECK 1

Negative cost means the measure saves money

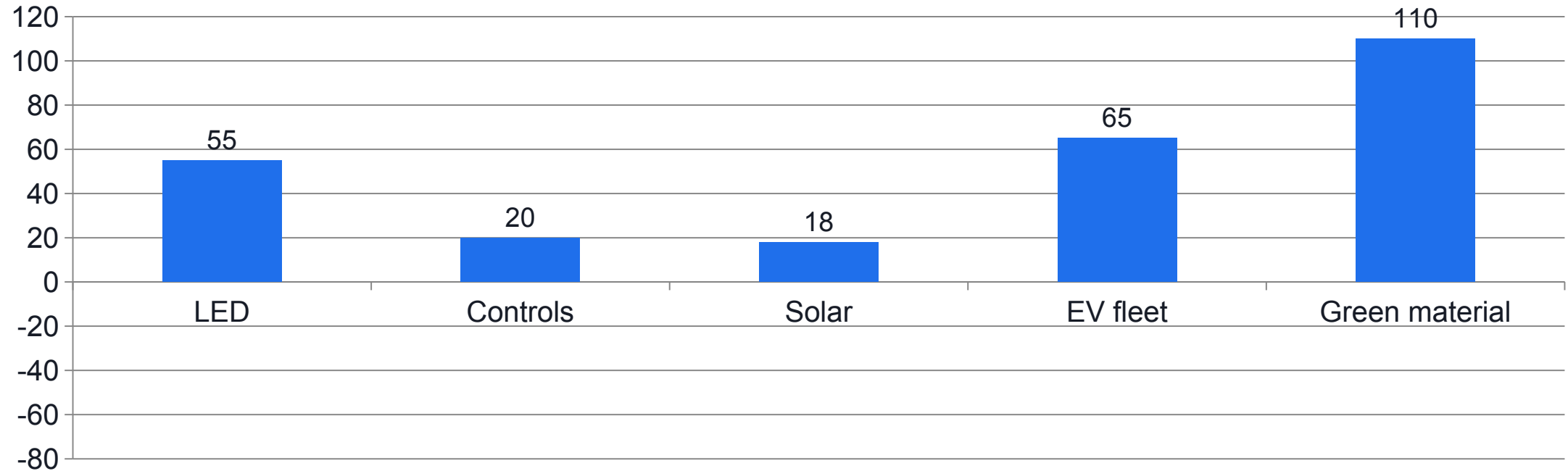
CHECK 2

Use consistent lifetime and discounting

CHECK 3

Do not ignore feasibility and dependencies

Illustrative Abatement Portfolio



WHAT THE DATA SHOWS

Negative-cost measures can help fund strategic measures with positive abatement cost.

Stress-Test the Variables That Matter

1

Energy price

Low, expected and high tariff cases.

2

Utilisation

Operating hours, throughput and occupancy.

3

Performance

Degradation, rebound and implementation shortfall.

4

Carbon value

Tax, internal carbon price or customer premium.

5

Discount rate

Financing and risk assumptions.

Two Ways to Price Carbon Internally

Shadow price

- Apply a hypothetical carbon value in investment cases
- Changes decisions without charging business units
- Good for stress-testing long-lived assets

Internal fee

- Charge emissions and recycle funds into reductions
- Creates a visible operating signal
- Needs governance and acceptance

LED Retrofit Decision

Illustrative teaching data: S\$120,000 capex; S\$38,000 annual saving; 72 tCO₂e avoided per year; 8-year life.

```
Payback = 120,000 ÷ 38,000 = 3.16 years  
Simple lifetime MAC = (120,000 - 8×38,000)  
÷ (8×72) = -S$319/tCO2e
```

CHECK 1

The project saves money and carbon

CHECK 2

Confirm maintenance, hours and baseline

CHECK 3

Use discounted cash flow for approval

From Standards to a Defensible Hotspot

1

Frame

Purpose, organisational boundary, operational boundary and period.

2

Quantify

Activity data × factor × GWP with controlled units.

3

Diagnose

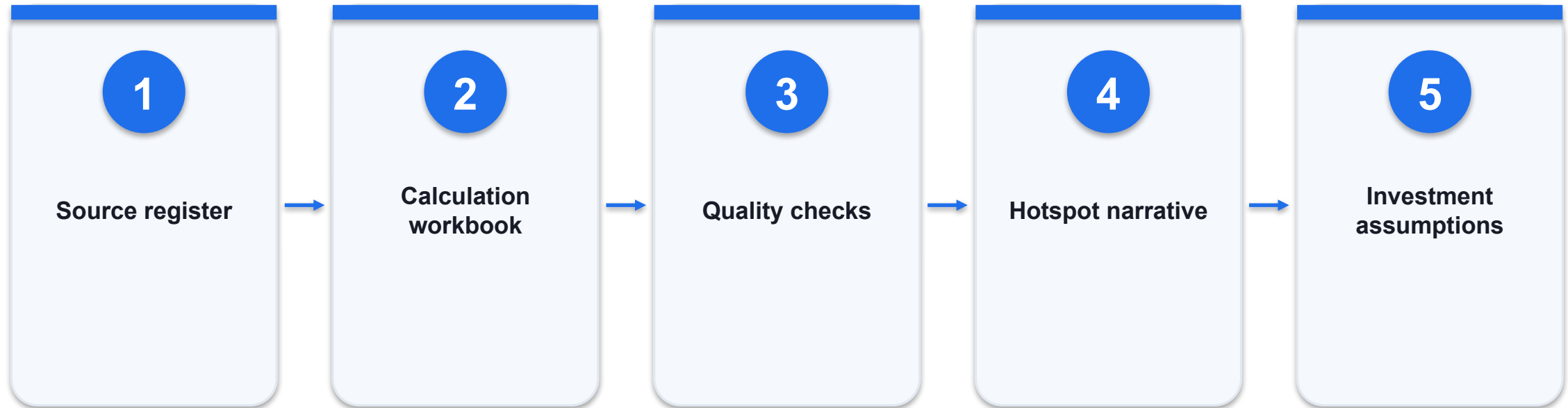
Rank magnitude, trend, risk and controllability.

4

Value

Combine tonnes, cash, risk and strategic fit.

Evidence Should Survive the Handover



TOPIC 1 · TAKEAWAY

Precision without completeness can produce a confidently wrong hotspot.

Screen broadly, then improve the data where decisions are material.

Map a Multi-Site Corporate Inventory

ACTIVITY 1

Define organisational and operational boundaries, classify sources and identify data gaps.

Define organisational and operational boundaries, classify sources and identify data gaps.

YOU'LL BUILD

Boundary memo, scope map and data-gap register

TOOLCHAIN

Case: Singapore retail and distribution group
Source lens: GHG Protocol Corporate Standard

DONE WHEN

Your recommendation is quantified, evidence-linked and explicit about uncertainty.

Use the SG Carbon Calculator as an Engagement Tool

ACTIVITY 2

Use personal-footprint screening responsibly, then translate behavioural insights into corporate data needs.

Use personal-footprint screening responsibly, then translate behavioural insights into corporate data needs.

YOU'LL BUILD

Calculator evidence, limitations note and engagement experiment

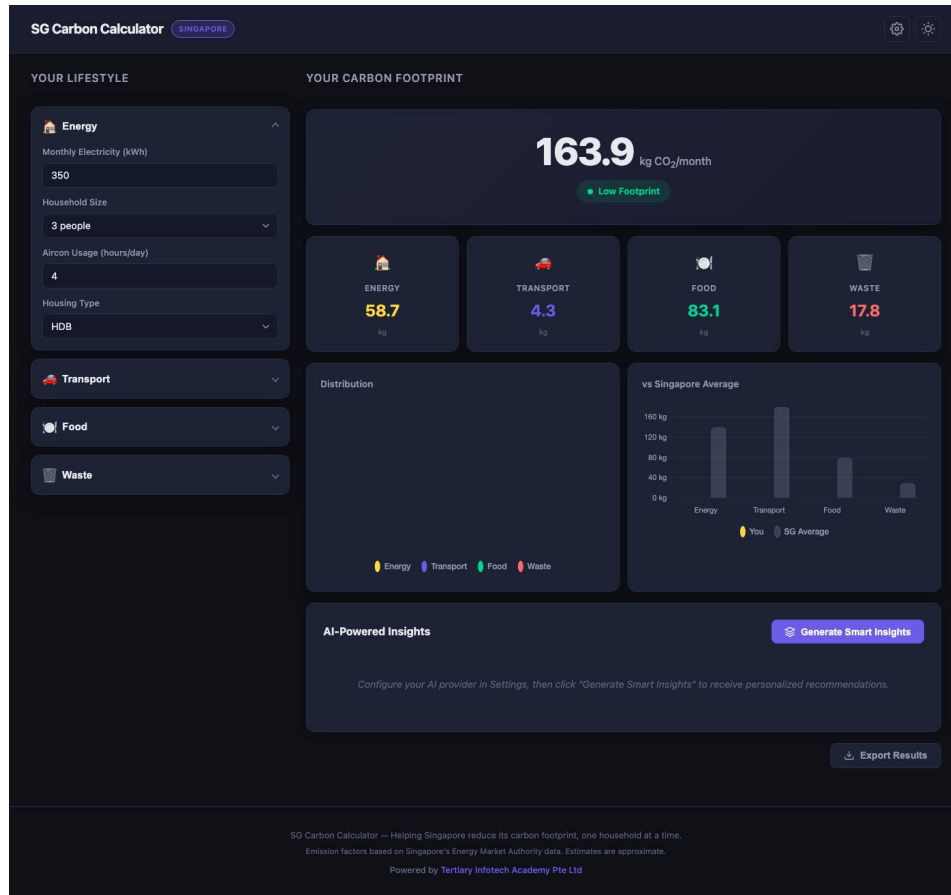
TOOLCHAIN

Case: Employee carbon-literacy campaign
Source lens: SG Carbon Calculator and UN ActNow

DONE WHEN

Your recommendation is quantified, evidence-linked and explicit about uncertainty.

SG Carbon Calculator: Use It for Screening, Not Inventory



RIGHT TOOL · RIGHT PURPOSE

- Explore how energy, transport, food and waste choices affect a personal screening result.
- Use fictional personas; avoid collecting unnecessary personal information.
- Translate insights into engagement hypotheses and operational data needs.
- Do not aggregate personal results into a corporate GHG inventory.

alfredang.github.io/sgcarboncalculator

Find Hotspots in a Farm-to-Table Value Chain

ACTIVITY 3

Calculate a screening footprint, rank hotspots and propose a data-improvement plan.

Calculate a screening footprint, rank hotspots and propose a data-improvement plan.

YOU'LL BUILD

Calculated footprint, Pareto chart and hotspot recommendation

TOOLCHAIN

Case: CP Foods-inspired teaching case
Source lens: SAP customer spotlight and GHG Protocol

DONE WHEN

Your recommendation is quantified, evidence-linked and explicit about uncertainty.

LUNCH BREAK

60 minutes

TOPIC 2 · K4 · K5 · K8 · A2

Prioritise Reduction Initiatives

Projection, technology, economics and transparent decisions

A Baseline Is a Managed Reference Point

1

Base year

Representative, reliable and relevant to the target.

2

Structural change

Acquisition, divestment or outsourcing may require recalculation.

3

Method change

Material factor or methodology changes may require restatement.

4

Growth

Separate absolute emissions from intensity and activity growth.

Start With the Business-as-Usual Case

Make growth, efficiency and factor assumptions explicit.

```
Future emissions = Future activity × Future  
emission factor × (1 – efficiency  
improvement)
```

CHECK 1

Build the no-action case first

CHECK 2

Avoid claiming reductions caused by grid decarbonisation as operational...

CHECK 3

Use compatible units and periods

Absolute and Intensity Targets Answer Different Questions

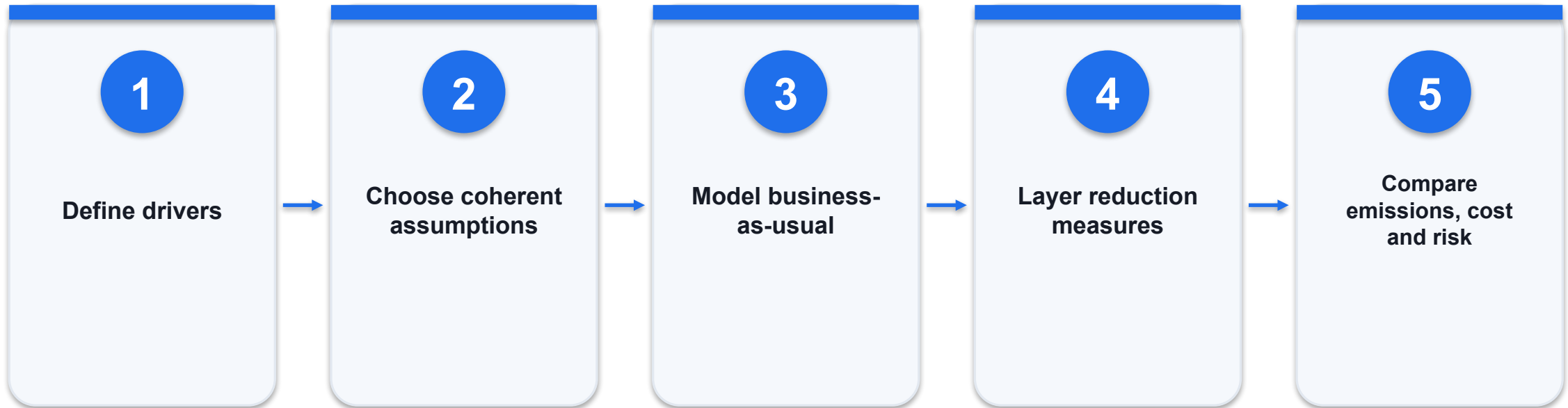
Absolute target

- Total tCO₂e falls against baseline
- Directly aligned to atmospheric outcome
- Can challenge a fast-growing business

Intensity target

- tCO₂e per unit of output or revenue
- Separates growth from efficiency
- Can improve while total emissions rise

Build Scenarios as Decision Stories



Use Ranges Where the Future Is Uncertain

1

Low case

Slower growth, lower adoption or conservative performance.

2

Expected case

Best current estimate with documented evidence.

3

High case

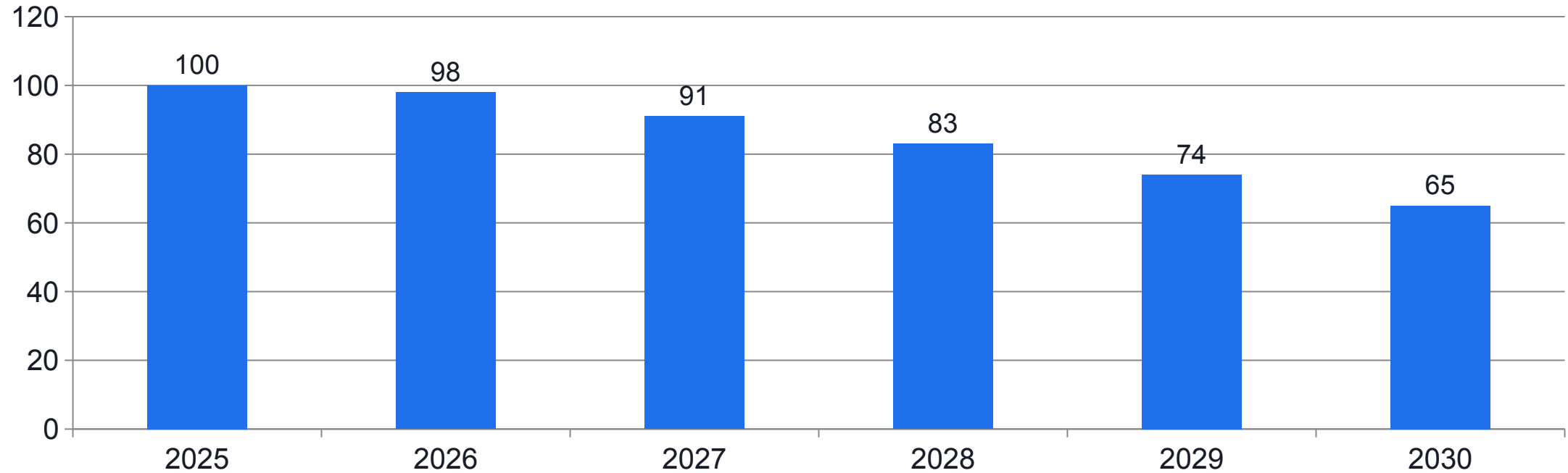
Faster growth, price shock or implementation upside.

4

Trigger

Define the observation that causes a plan revision.

Illustrative 2030 Emissions Path



WHAT THE DATA SHOWS

A target path becomes actionable only when initiatives, owners and milestones explain each step down.

Explain Why Emissions Changed

Separate operational performance from structural and market effects.

$\Delta \text{ Emissions} = \text{Activity effect} + \text{Efficiency effect} + \text{Factor effect} + \text{Structural effect}$

CHECK 1

Activity effect captures volume or growth

CHECK 2

Efficiency effect captures operational improvement

CHECK 3

Factor effect captures energy or supplier decarbonisation

Avoid the Rebound Trap

1

Efficiency gain

Less energy per service unit.

2

Behaviour response

Lower cost can increase demand or operating hours.

3

Net result

Gross saving minus additional consumption.

4

Control

Track both intensity and absolute activity.

Electrifying a Singapore Delivery Fleet

Illustrative: 20 vans × 35,000 km/year; diesel van 0.25 kg CO₂e/km; EV 0.20 kWh/km; grid factor 0.402 kg CO₂/kWh.

```
Diesel = 175 tCO2e/year  
EV electricity = 56.3 tCO2e/year  
Screening reduction = 118.7 tCO2e/year
```

CHECK 1

Add vehicle and battery lifecycle where material

CHECK 2

Model future grid factors separately

CHECK 3

Validate actual route, charging and payload data

Reduce Before You Compensate

1

Avoid

Remove unnecessary demand, travel, material or process.

2

Reduce

Improve efficiency and yield.

3

Substitute

Use lower-carbon energy, material or transport.

4

Engage

Work with suppliers, customers and employees.

5

Neutralise residuals

Use credible removals only after deep reductions.

Energy Levers: Measure, Control, Replace

1

Measure

Sub-meter material loads and normalise for production.

2

Control

Schedules, set-points, maintenance and automation.

3

Upgrade

Efficient motors, chillers, compressed air and heat recovery.

4

Replace

Renewable electricity and electrified heat where feasible.

Operational Excellence Is Carbon Management

1

Yield

Prevent scrap and rework.

2

Flow

Reduce waiting, idle loads and unnecessary movement.

3

Maintenance

Keep assets at designed efficiency.

4

Planning

Match batches, routes and capacity to demand.

5

Design

Remove carbon before it is locked into the process.

Supplier Engagement Needs a Data Ladder

1

Screen

Use secondary factors to find material categories.

2

Request

Ask material suppliers for product or process data.

3

Validate

Check boundary, method, period and assurance.

4

Collaborate

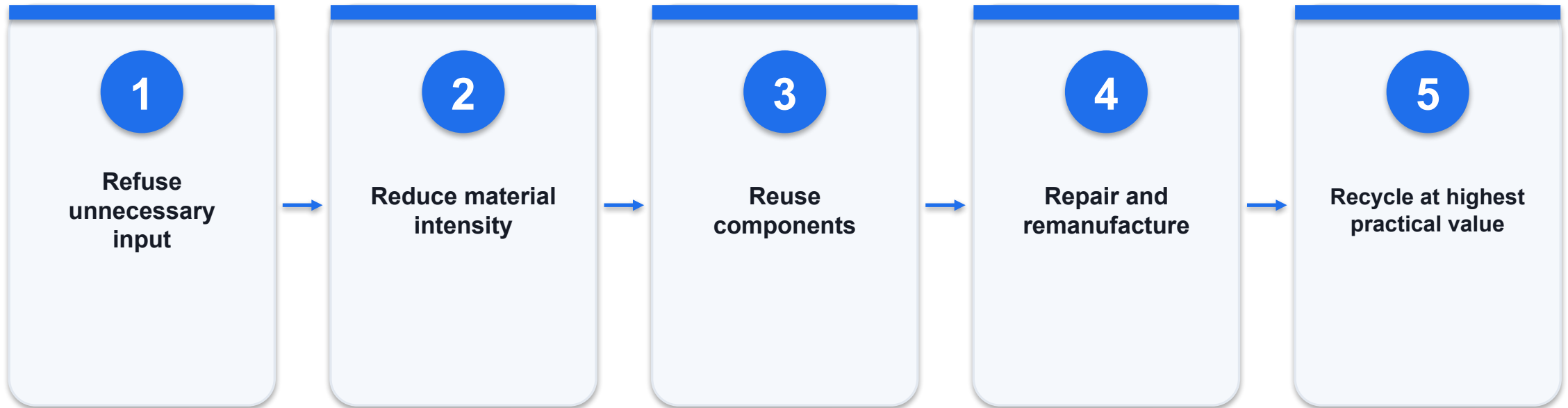
Co-design lower-carbon specifications and logistics.

5

Contract

Embed evidence and performance expectations.

Circular Design Changes the Carbon Equation



Industry 4.0 Can Help—and Create New Risks

Opportunity

- IoT enables real-time energy and process data
- Analytics detects anomalies and predicts failures
- ERP connects carbon to materials, orders and suppliers

Implementation risk

- High integration cost and interoperability gaps
- Data security and ownership concerns
- Automation can scale poor-quality data

What an Enterprise Footprint Platform Connects

1

Transactions

Materials, purchase orders, production and logistics.

2

Factors

Versioned libraries matched to geography and period.

3

Flows

Energy and material movement across products and sites.

4

Results

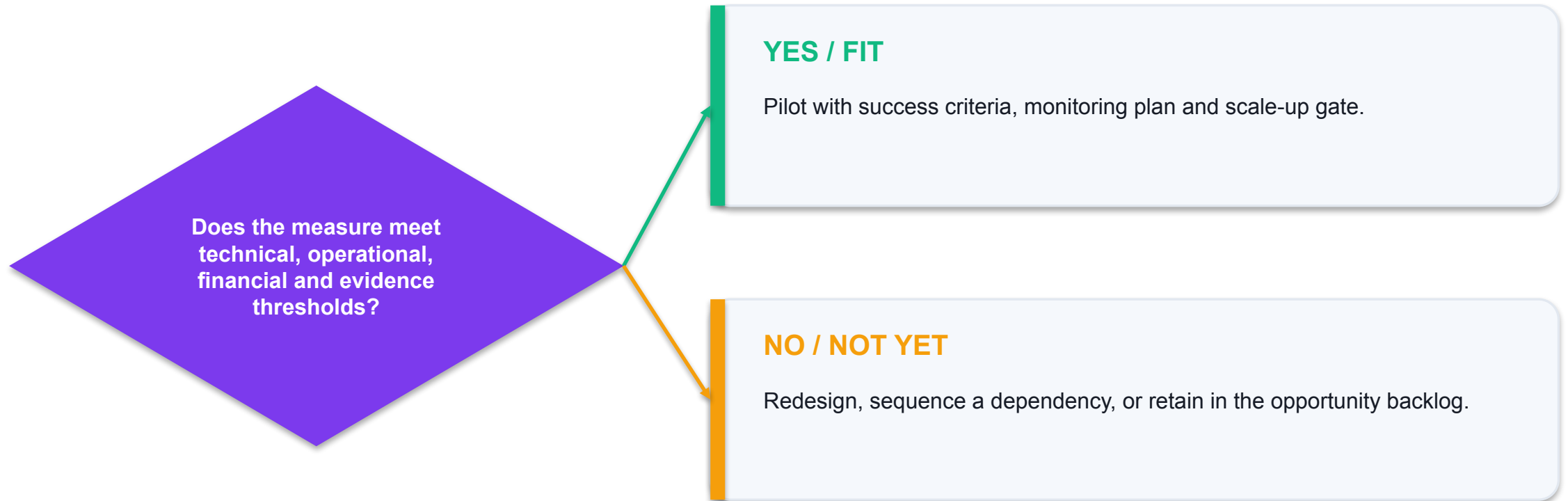
Corporate and product footprints with drill-down.

5

Action

Hotspots integrated into operational decisions.

Is the Technology Ready for This Site?



A promising technology is not yet a feasible initiative until site constraints and implementation risk are tested.

Design a Pilot That Can Be Judged

1

Baseline

Comparable pre-intervention period and operating conditions.

2

Intervention

Specific asset, control or behaviour change.

3

Metric

Energy, output, emissions, cost and quality guardrail.

4

Duration

Long enough to capture representative variation.

5

Decision rule

Scale, modify or stop threshold agreed in advance.

A Portfolio Needs More Than One Score

1

Impact

Annual and cumulative tCO₂e.

2

Economics

NPV, payback and abatement cost.

3

Feasibility

Technology maturity, resources and disruption.

4

Strategic fit

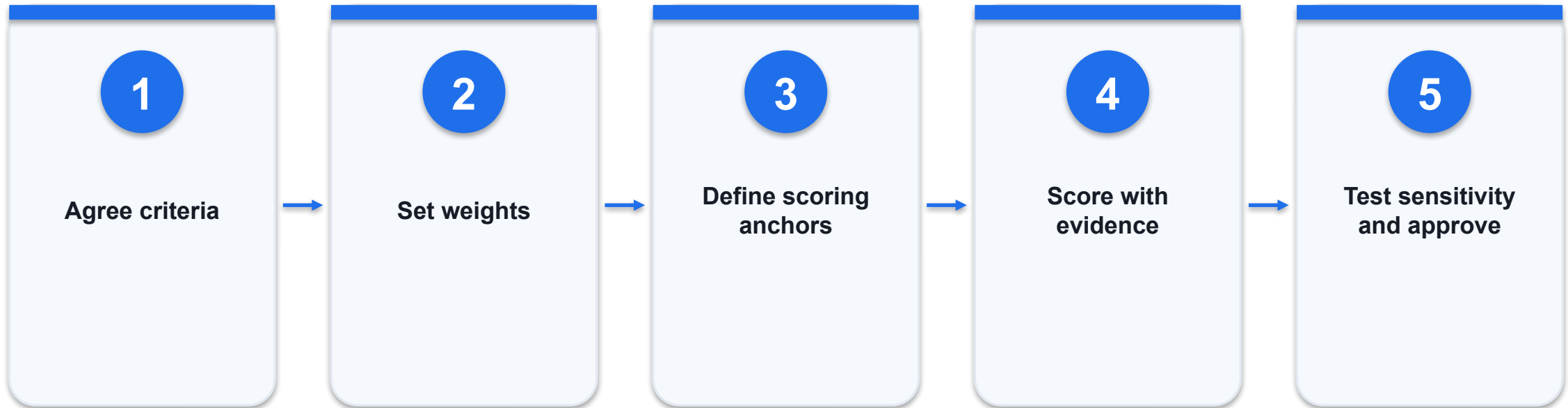
Customer, compliance and resilience value.

5

Confidence

Data quality and dependency risk.

A Transparent Multi-Criteria Decision Process



Make Trade-Offs Visible

Use a common scale and documented evidence.

Weighted score = $\sum$ (criterion score × criterion weight)

CHECK 1

Weights must total 100%

CHECK 2

Use behavioural anchors for 1–5 scores

CHECK 3

Do not let the formula replace governance

No-Regret Measures and Strategic Bets

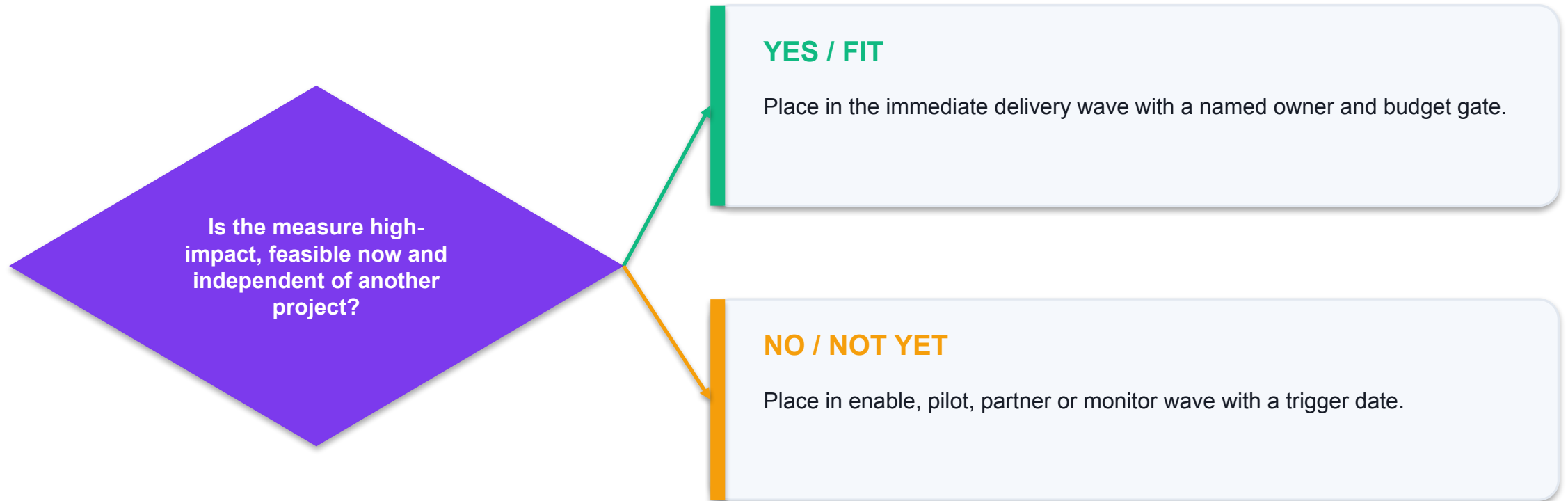
No-regret

- Strong economics and proven technology
- Low dependency and quick implementation
- Build momentum and fund later work

Strategic bet

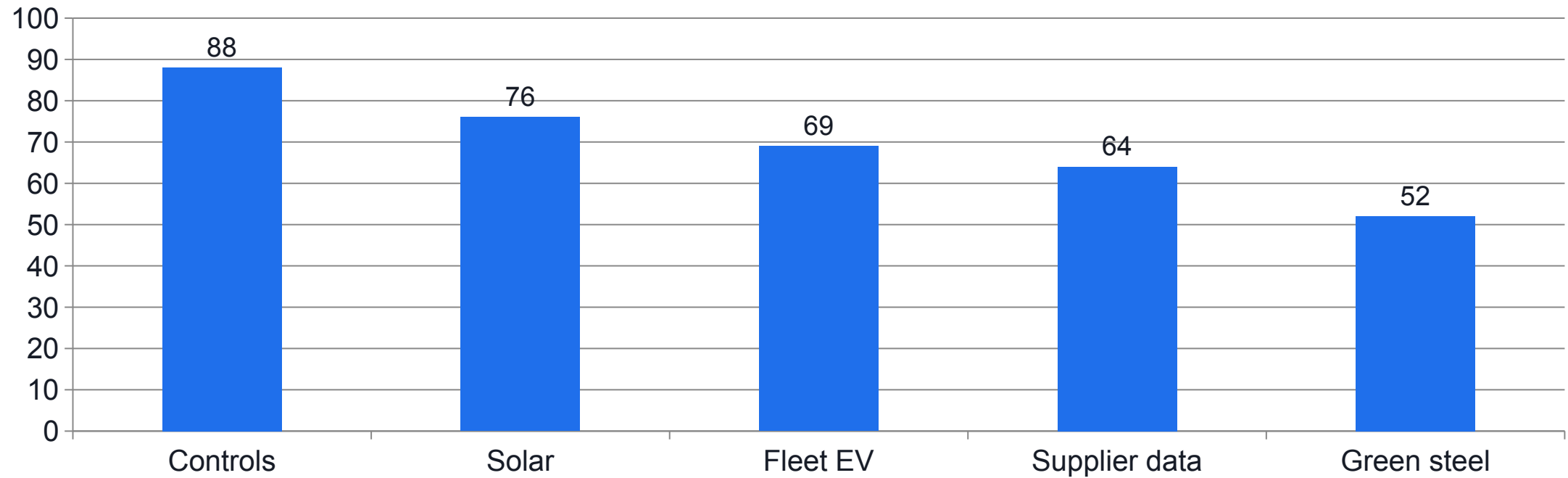
- Higher uncertainty or capital need
- Enables deep reductions or future compliance
- Use staged gates and learning milestones

What Should Start First?



Sequence dependencies: data platform before optimisation, charging before fleet conversion, specification before supplier tender.

Illustrative Initiative Scores



WHAT THE DATA SHOWS

A score supports discussion; the approval record must still explain constraints, dependencies and dissent.

Counter Common Decision Biases

1

Payback bias

Include lifecycle value and residual risk.

2

Pet-project bias

Require comparable evidence and scoring anchors.

3

Precision bias

Show ranges and confidence ratings.

4

Silo bias

Include operations, finance, procurement and users.

5

Offset bias

Apply the reduction hierarchy first.

Portfolio Choice Under a S\$500k Cap

Illustrative measures: controls S\$80k/160 t; solar S\$280k/210 t; EV pilot S\$140k/90 t; green material S\$220k/300 t.

Possible Wave 1: controls + solar + EV
pilot = S\$500k and 460 tCO₂e/year screened
benefit

CHECK 1

Green material has high impact but needs
procurement validation

CHECK 2

Check double counting and dependencies

CHECK 3

Retain a reserve for measurement and
commissioning

Prioritisation Is a Portfolio Conversation

1

Project

Model baseline, intervention and uncertainty.

2

Prove

Test technical and operating feasibility.

3

Compare

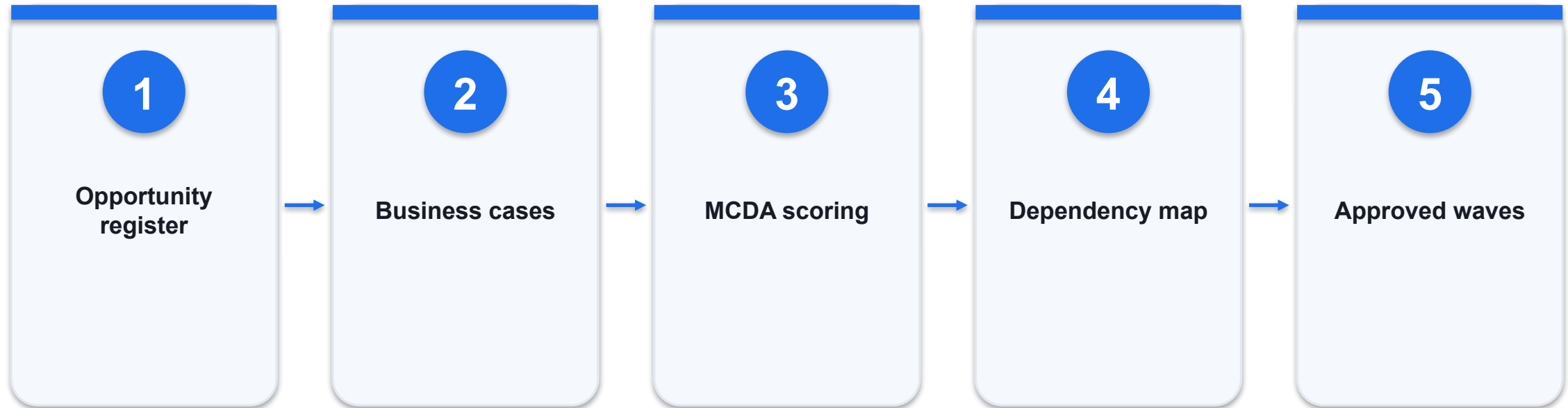
Use carbon, cash, risk, strategy and confidence.

4

Sequence

Balance no-regret delivery with deep-reduction enablers.

From Long List to Funded Roadmap



TOPIC 2 · TAKEAWAY

The best portfolio is credible enough to fund and flexible enough to learn.

Use scenarios, pilots and stage gates to turn uncertainty into managed evidence.

Model a Singapore Fleet Transition

ACTIVITY 4

Build business-as-usual and reduction scenarios for diesel, route optimisation and fleet electrification.

Build business-as-usual and reduction scenarios for diesel, route optimisation and fleet electrification.

YOU'LL BUILD

2030 scenario model, assumptions register and recommendation

TOOLCHAIN

Case: LionCity Logistics Source lens: EMA grid statistics and GHG Protocol

DONE WHEN

Your recommendation is quantified, evidence-linked and explicit about uncertainty.

Screen a Smart-Factory Carbon Initiative

ACTIVITY 5

Assess the technical, operational, data and financial feasibility of IoT/ERP-enabled energy optimisation.

Assess the technical, operational, data and financial feasibility of IoT/ERP-enabled energy optimisation.

YOU'LL BUILD

Feasibility scorecard and pilot charter

TOOLCHAIN

Case: Matsumoto Precision-inspired teaching case
Source lens: SAP customer spotlight and Yurtay (2025)

DONE WHEN

Your recommendation is quantified, evidence-linked and explicit about uncertainty.

Prioritise a Reduction Portfolio

ACTIVITY 6

Combine MACC, weighted criteria, dependencies and budget constraints into a transparent portfolio.

Combine MACC, weighted criteria, dependencies and budget constraints into a transparent portfolio.

YOU'LL BUILD

MACC, MCDA matrix, funded roadmap and decision record

TOOLCHAIN

Case: WITTE Automotive-inspired teaching case
Source lens: SAP customer spotlight, CarbonChain and IMD

DONE WHEN

Your recommendation is quantified, evidence-linked and explicit about uncertainty.

SHORT BREAK

15 minutes

TOPIC 3 · K2 · K6 · K9 · A3

Develop the Reduction Plan

EMS governance, action planning, Singapore requirements and review

An EMS Turns Intent Into Repeatable Control

1

Context

Interested parties, risks, opportunities and obligations.

2

Leadership

Policy, accountability and resources.

3

Planning

Aspects, objectives, targets and actions.

4

Support & operation

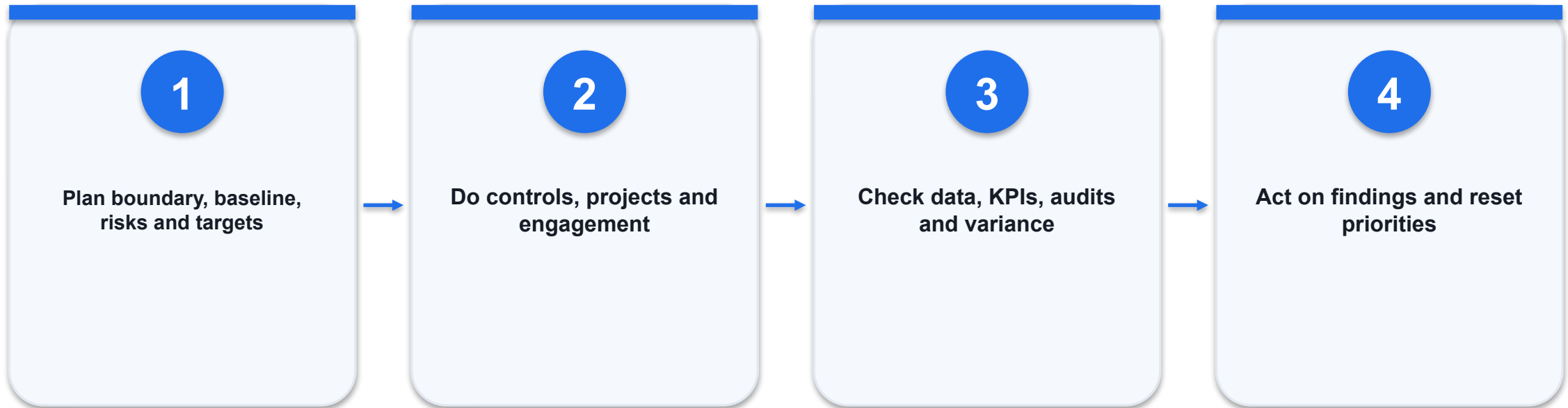
Competence, communication and operational controls.

5

Evaluation & improvement

Monitor, audit, review and correct.

Carbon Management Through PDCA



Ownership and Accountability Are Different

Process owner

- Controls the activity and operational data
- Implements agreed actions
- Explains performance variance

Carbon / GHG manager

- Maintains method, inventory and controls
- Coordinates reporting and assurance
- Escalates quality and compliance issues

Build Controls Around the Data Journey

1

Capture

Authorised sources and complete populations.

2

Transform

Locked factors, units and calculation rules.

3

Review

Threshold checks and independent approval.

4

Change

Version control and recalculation log.

5

Retain

Evidence repository and retention period.

The Plan Needs People Who Can Operate It

1

Executives

Approve ambition, capital and risk appetite.

2

Finance

Validate business cases and reporting controls.

3

Operations

Own sources, projects and root causes.

4

Procurement

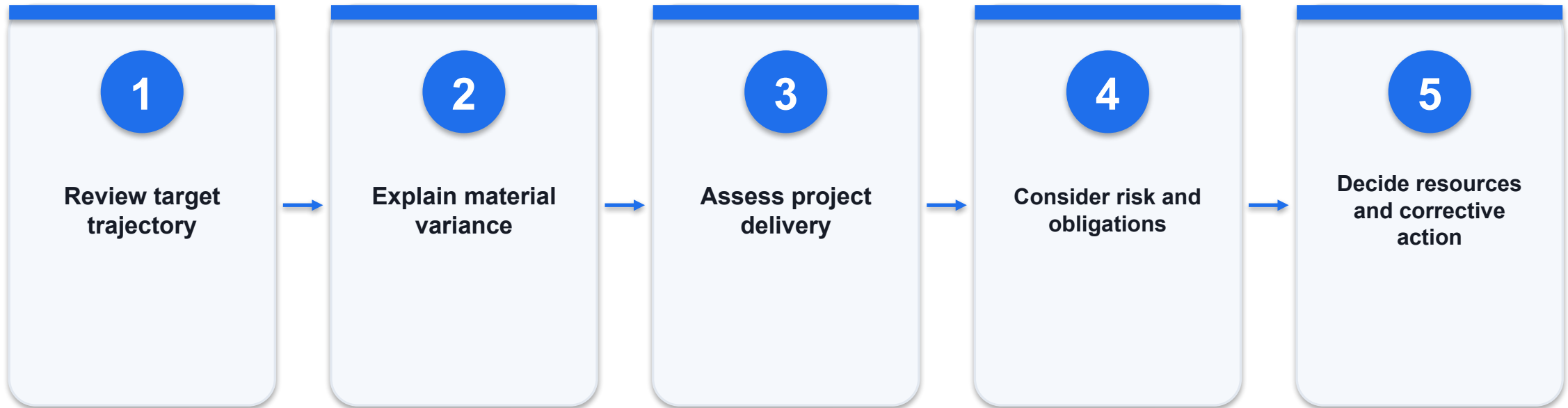
Engage suppliers and embed specifications.

5

Data / IT

Maintain architecture, access and quality.

A Useful Review Drives Decisions



Audit the System, Not Only the Number

1

Boundary

Is it complete and consistently applied?

2

Evidence

Can samples be traced to source?

3

Calculation

Are factors, units and formulas controlled?

4

Action

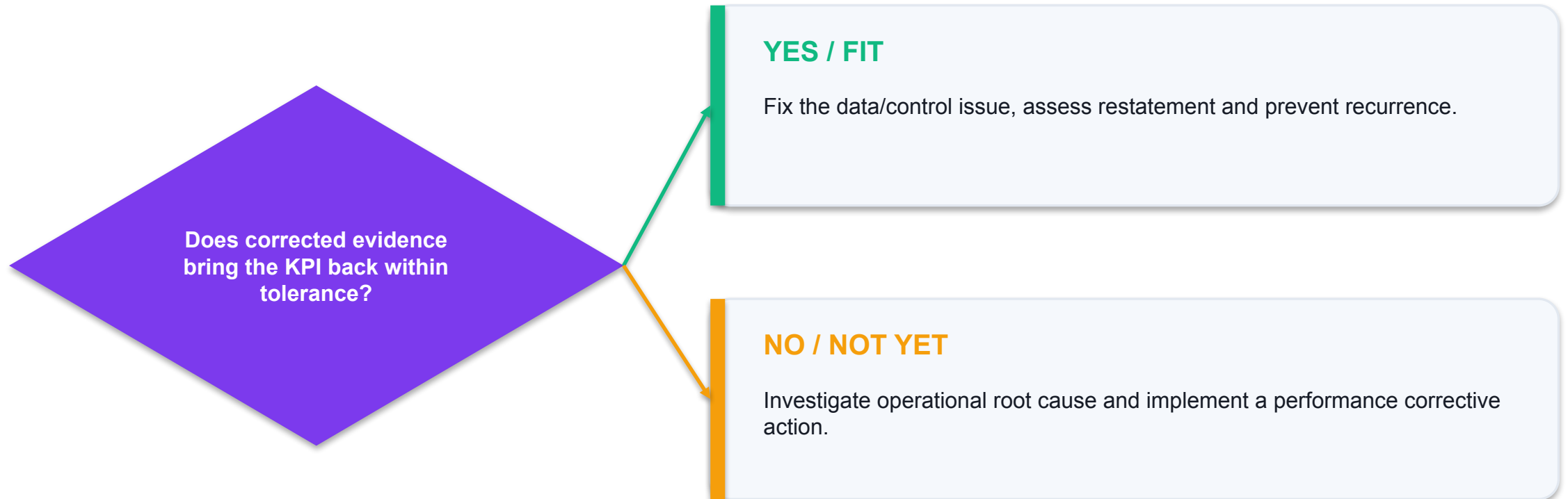
Are corrective actions owned and closed?

5

Communication

Are claims supported and balanced?

Is the Variance a Data Error or Performance Gap?



Do not hide a missed target by changing the data; do not launch a project to fix a calculation error.

K2 · TAKEAWAY

A footprint becomes manageable when it has owners, controls and review rhythms.

The inventory is an operating process—not an annual spreadsheet event.

A Credible Reduction Plan Has Eight Parts

1

1 · Boundary and baseline

What is covered and measured.

2

2 · Hotspots

Where emissions concentrate and why.

3

3 · Target

Outcome, year, scope and trajectory.

4

4 · Initiatives

Measures, savings and dependencies.

5

5 · Roadmap

Sequence, milestones and resources.

6

6 · Governance

Owners and decision rights.

7

7 · Measurement

KPIs, evidence and review.

8

8 · Risk & communication

Uncertainty, claims and escalation.

A Target Needs Scope and Substance

Weak target

- Reduce emissions significantly
- No base year or boundary
- No trajectory or owner

Decision-ready target

- 35% absolute Scope 1+2 reduction by 2030 from 2025
- Boundary and recalculation policy stated
- Annual trajectory and accountable executive

Sequence the Reduction Plan



Connect Outcome, Driver and Delivery Metrics

1

Outcome

Total tCO₂e and tCO₂e per functional unit.

2

Driver

kWh/unit, fuel/km, kg material/unit, waste yield.

3

Delivery

Project milestone, installed capacity, supplier coverage.

4

Quality

Primary-data share, estimate share and control exceptions.

Make Failure Modes Visible Before Launch

1

Delivery

Delay, skills, vendor or permit.

2

Performance

Savings lower than modelled.

3

Commercial

Tariff, price, demand or finance.

4

Data

Missing evidence or incompatible systems.

5

Claims

Overstatement, double counting or unclear offsets.

Reduction and Offset Are Not Interchangeable

Operational reduction

- Changes emissions within the inventory or value chain
- Traceable to a measure and baseline
- Supports the reduction trajectory

Offset / credit

- Finances a reduction or removal outside the inventory
- Requires quality and retirement evidence
- Disclose separately from gross emissions

A Responsible Claim Shows the Full Picture

1

Boundary

Which entities, scopes and period?

2

Metric

Gross, net, absolute or intensity?

3

Method

Which standard, factors and recalculations?

4

Progress

Actual result versus trajectory.

5

Limitations

Estimates, uncertainty and exclusions.

Translate a 35% Target Into Annual Milestones

Illustrative 2025 baseline: 10,000 tCO₂e. 2030 target: 6,500 tCO₂e.

Required reduction = 3,500 tCO₂e
Linear planning guide = 700 tCO₂e/year
average

CHECK 1

Actual roadmap need not be linear

CHECK 2

Link each step change to initiatives

CHECK 3

Track business-as-usual growth separately

Budget the Enablers, Not Only the Hardware

1

Measurement

Meters, factors, assurance and data engineering.

2

People

Project management, training and supplier engagement.

3

Change

Communications, workflow redesign and incentives.

4

Contingency

Performance shortfall and dependency risk.

5

Maintenance

Commissioning and sustained optimisation.

Every Action Needs a Complete Contract

1

What

Specific intervention and boundary.

2

Why

Hotspot and expected outcome.

3

Who

Accountable owner and support roles.

4

When

Start, milestone, completion and review.

5

How much

Budget, resources and quantified benefit.

6

Evidence

KPI, source and verification method.

Make Objectives SMARTER

A target should guide action and adaptation.

Specific · Measurable · Achievable ·
Relevant · Time-bound · Evaluated ·
Reviewed

CHECK 1

Use an outcome metric and at least one driver metric

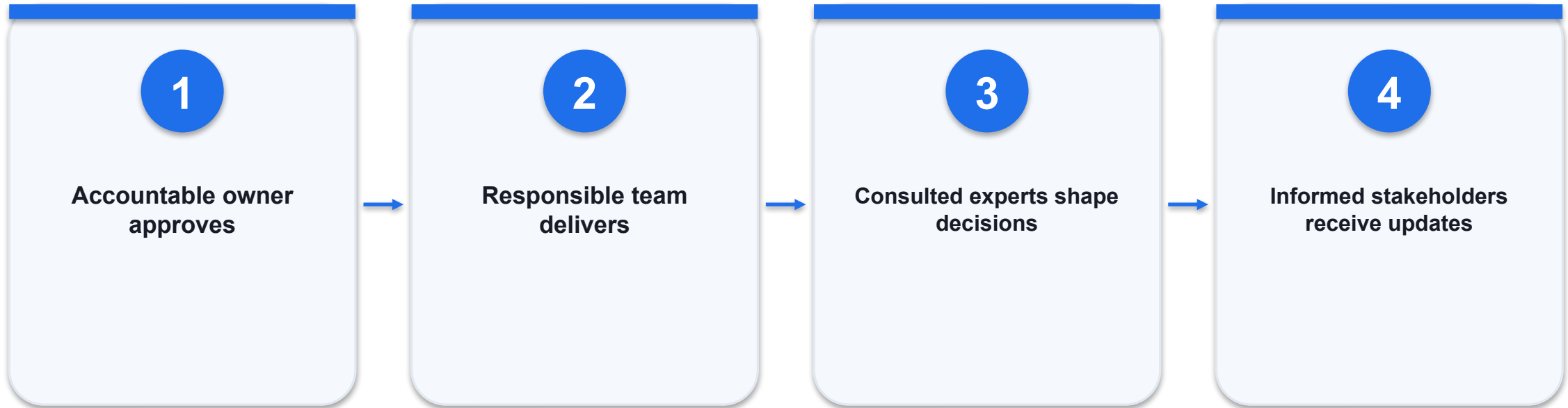
CHECK 2

Name the evidence source and review cadence

CHECK 3

Predefine escalation and revision conditions

Clarify Roles Before Delivery Starts



Use a Single Initiative Register

1

Identity

Initiative ID, source, site and owner.

2

Value

Capex, savings, tCO₂e and confidence.

3

Delivery

Stage, milestone, dependency and issue.

4

Decision

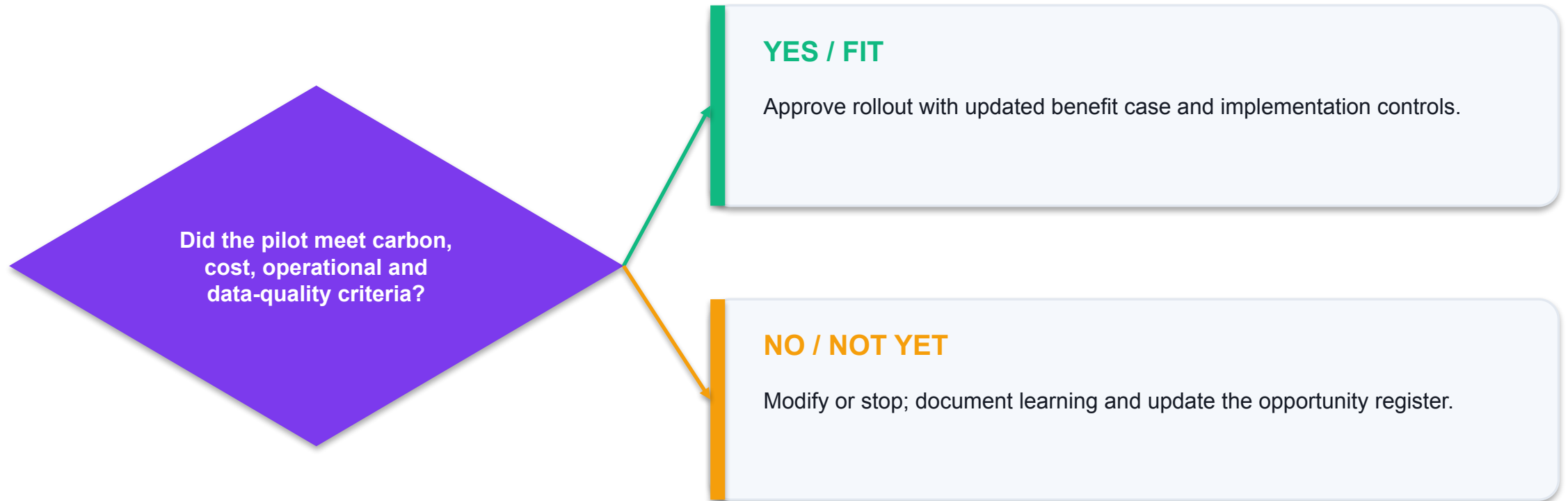
Approval, gate and rationale.

5

Evidence

Baseline, M&V method and realised result.

Should the Initiative Scale?



Stage gates protect capital while preserving learning from uncertainty.

Measure Realised Savings Fairly

1

Baseline model

Representative pre-project performance.

2

Adjustment

Production, weather, occupancy or route mix.

3

Boundary

Avoid leakage and double counting.

4

Persistence

Check that savings continue after launch.

5

Approval

Independent review of material results.

Know the Current Singapore Thresholds

1

Reportable facility

Annual direct emissions at or above 2,000 tCO₂e.

2

Taxable facility

Annual direct emissions at or above 25,000 tCO₂e.

3

2026–2027 tax rate

S\$45 per tCO₂e.

4

Toward 2030

Government has indicated S\$50–S\$80 per tCO₂e by 2030.

5

Scope

Industrial facilities in specified sectors; verify current CPA requirements.

Reportable and Taxable Facilities

≥2,000 tCO₂e direct

- Register as reportable where applicable
- Submit annual Emissions Report by 30 June following year
- Appoint required representatives and manager

≥25,000 tCO₂e direct

- Also subject to carbon tax
- Monitoring Plan and verified Emissions Report
- Stronger controls and verification requirements

Screen the 2026 Carbon-Tax Exposure

Illustrative taxable emissions: 30,000 tCO₂e. 2026 rate: S\$45/tCO₂e.

Gross screened liability = 30,000 × S\$45 =
S\$1,350,000

CHECK 1

Use verified reportable emissions and current law for filing

CHECK 2

Account for eligible allowances only with competent advice

CHECK 3

Use the result to stress-test investments—not as the inventory boundary

A Plan Is a Managed System

1

Govern

Owners, controls, competence and management review.

2

Plan

Boundary, target, initiatives, roadmap and resources.

3

Deliver

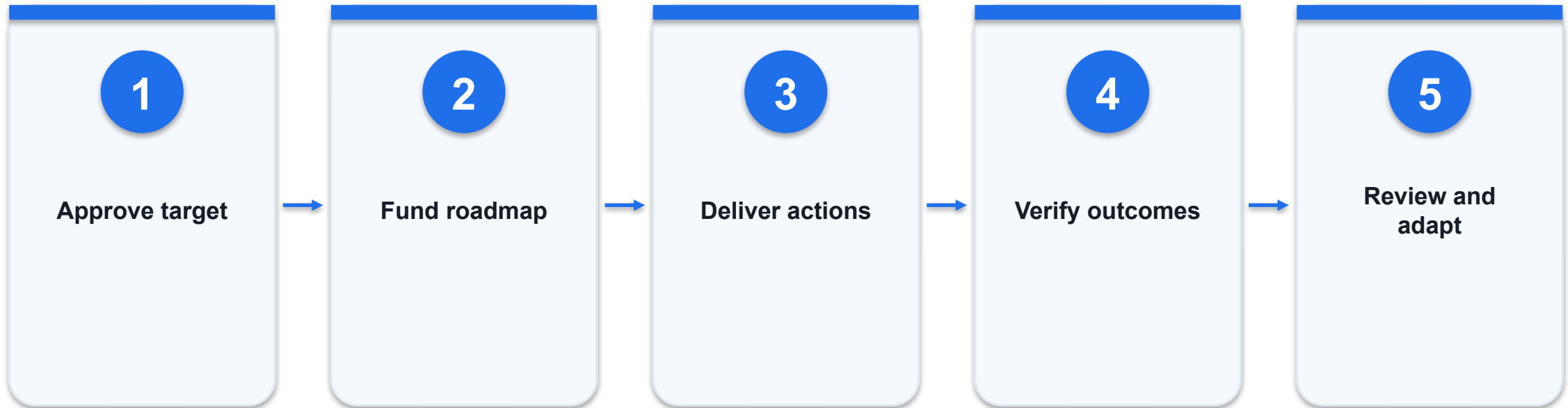
Action contracts, stage gates and dependencies.

4

Learn

Measurement, audit, corrective action and transparent claims.

From Target to Continual Improvement



TOPIC 3 · TAKEAWAY

The reduction plan is credible when someone can execute it, finance it and verify it.

Targets become performance only through governance and evidence.

Prepare a Singapore MR and Carbon-Tax Readiness Plan

ACTIVITY 7

Screen regulatory status, data controls, reporting obligations and carbon-price exposure.

Screen regulatory status, data controls, reporting obligations and carbon-price exposure.

YOU'LL BUILD

Obligations matrix, control plan and screened tax exposure

TOOLCHAIN

Case: Jurong specialty-chemicals facility
Source lens: NEA Carbon Tax and MR requirements

DONE WHEN

Your recommendation is quantified, evidence-linked and explicit about uncertainty.

Design an Employee ActNow Programme

ACTIVITY 8

Convert selected personal actions into a privacy-respecting corporate engagement programme with measurable outcomes.

Convert selected personal actions into a privacy-respecting corporate engagement programme with measurable outcomes.

YOU'LL BUILD

Programme canvas, measurement plan and communication copy

TOOLCHAIN

Case: Singapore professional-services firm
Source lens: United Nations ActNow

DONE WHEN

Your recommendation is quantified, evidence-linked and explicit about uncertainty.

Build the Board-Ready Carbon Reduction Plan

ACTIVITY 9

Integrate hotspot analysis, prioritisation, governance and action planning into a defensible 2030 plan.

Integrate hotspot analysis, prioritisation, governance and action planning into a defensible 2030 plan.

YOU'LL BUILD

Ten-slide board pack plus action register and management-review dashboard

TOOLCHAIN

Case: Lion City Precision Pte Ltd
Source lens: Integrated course case using GHG Protocol, ISO/EMS and Singapore context

DONE WHEN

Your recommendation is quantified, evidence-linked and explicit about uncertainty.



EVIDENCE BASE

Standards, Regulation and Further Reading

Verify current requirements before external reporting

Source Register

1

GHG Protocol Corporate Standard

ghgprotocol.org

2

United Nations ActNow — Ten Actions

un.org

3

NEA — Singapore Carbon Tax

nea.gov.sg

4

NEA — Measurement and Reporting Requirements

nea.gov.sg

Source Register

1

EMA — Singapore Energy Statistics, Chapter 2

ema.gov.sg

2

SAP Sustainability Footprint Management

sap.com

3

IMD — Carbon Management

imd.org

4

Yurtay (2025), Industry 4.0 and ERP Systems

mdpi.com

Source Register

1

Synesgy — Carbon Footprint Guide

synesgy.com

2

Green Earth — Carbon Footprints

green.earth

3

CarbonChain — Reducing Company Footprints

carbonchain.com

4

SG Carbon Calculator

github.com

 WRAP-UP

From Footprint to Managed Reduction

Integrate measurement, decisions and delivery

What You Can Now Do

1

LO1

Analyse emission hotspots with defensible boundaries and data.

2

LO2

Prioritise a portfolio using scenarios, feasibility, economics and transparent criteria.

3

LO3

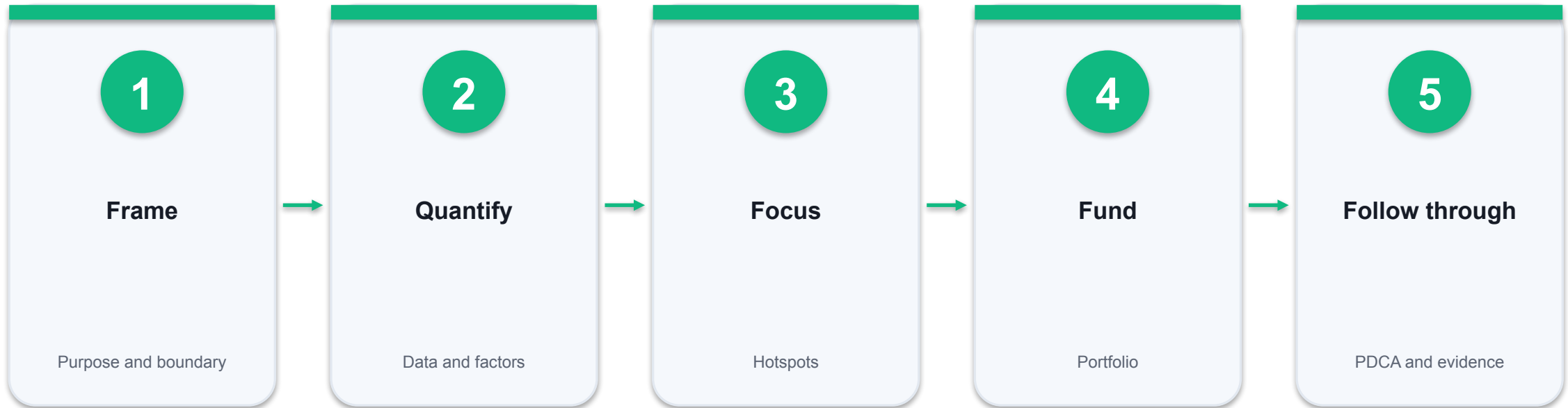
Develop an executable reduction plan with targets, governance and measurement.

4

Professional judgement

State assumptions, uncertainty and limitations without hiding them.

Your Practical Playbook



Recommended Courses

1

WSQ Sustainability Reporting and Assurance

2

WSQ Environmental, Social and Governance (ESG) Essentials

3

WSQ Data Analytics for Sustainability

4

WSQ Business Process Automation with Power Automate

5

WSQ Business Innovation with Artificial Intelligence

Support

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Website

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LMS

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Assessment

1

Written Assessment

9 SAQs · 1 hour · open book

2

Case Study

3 applied questions · 1 hour · open book

3

Digital attendance

Complete the assessment attendance event

4

Submission

Follow the assessor's stated LMS or hard-copy process

Assessment Flow



Digital Attendance (Mandatory)

1

Complete every event

AM, PM and Assessment attendance

2

Verify success

Do not assume a failed scan is recorded

3

Minimum 75%

Attendance eligibility threshold

4

Ask for help

Resolve issues with the trainer immediately

CARBON MANAGEMENT FOR SUSTAINABILITY

Thank You

Measure what matters. Reduce what matters most. Prove what changed.